# EXPERIMENT – 3

**AIM:** To design and implement an Employee–Department–Project database schema and perform SQL queries using Selection, Projection, Aggregate Functions, GROUP BY, HAVING, CASE expressions, and ORDER BY.

## SOURCE CODE:

### DATABASE CREATING

```
CREATE DATABASE EmployeeManagement;
USE EmployeeManagement;
```

### TABLE CREATING

-- Department Table

```
CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL,
    location VARCHAR(50)
);
```

-- Employee Table

```
CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50) NOT NULL,
    gender CHAR(1),
    salary DECIMAL(10,2),
    hire_date DATE,
    dept_id INT,
    FOREIGN KEY (dept_id)
        REFERENCES Department(dept_id)
);
```

-- Project Table

```
CREATE TABLE Project (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(100) NOT NULL,
    budget DECIMAL(12,2),
    dept_id INT,
    FOREIGN KEY (dept_id)
        REFERENCES Department(dept_id)
);
```

-- Employee-Project Relationship

```
CREATE TABLE Employee_Project (
    emp_id INT,
    project_id INT,
    hours_worked INT,

    PRIMARY KEY (emp_id, project_id),

    FOREIGN KEY (emp_id)
        REFERENCES Employee(emp_id),

    FOREIGN KEY (project_id)
        REFERENCES Project(project_id)
);
```

**INSERTING DEPARTMENTS**

```
INSERT INTO Department (dept_id, dept_name, location)
VALUES
(1, 'IT', 'Delhi'),
(2, 'HR', 'Mumbai'),
(3, 'Finance', 'Bangalore'),
(4, 'Marketing', 'Pune'),
(5, 'Research', 'Hyderabad');
```

**INSERTING 30 EMPLOYEES**

```
INSERT INTO Employee
(emp_id, emp_name, gender, salary, hire_date, dept_id)
VALUES
(101, 'Aarav Sharma', 'M', 75000, '2021-01-15', 1),
(102, 'Ananya Singh', 'F', 68000, '2022-03-10', 1),
(103, 'Rohan Verma', 'M', 82000, '2020-07-20', 1),
(104, 'Priya Gupta', 'F', 72000, '2021-11-05', 1),
(105, 'Karan Mehta', 'M', 95000, '2019-06-18', 1),
(106, 'Sneha Kapoor', 'F', 64000, '2023-01-12', 1),
(107, 'Rahul Kumar', 'M', 55000, '2022-05-15', 2),
(108, 'Neha Sharma', 'F', 62000, '2021-09-22', 2),
(109, 'Amit Jain', 'M', 58000, '2020-04-11', 2),
(110, 'Pooja Verma', 'F', 67000, '2023-02-19', 2),
(111, 'Vikas Singh', 'M', 60000, '2022-08-01', 2),
(112, 'Riya Agarwal', 'F', 71000, '2019-12-10', 2),
(113, 'Arjun Patel', 'M', 88000, '2020-01-25', 3),
(114, 'Meera Shah', 'F', 92000, '2019-08-14', 3),
(115, 'Suresh Rao', 'M', 76000, '2021-06-30', 3),
(116, 'Kavya Iyer', 'F', 85000, '2022-10-17', 3),
(117, 'Manish Gupta', 'M', 73000, '2023-03-21', 3),
```

```
(118, 'Nisha Reddy', 'F', 90000, '2020-11-09', 3),
(119, 'Aditya Joshi', 'M', 70000, '2021-02-28', 4),
(120, 'Simran Kaur', 'F', 66000, '2022-04-15', 4),
(121, 'Mohit Bansal', 'M', 74000, '2020-09-12', 4),
(122, 'Isha Malhotra', 'F', 81000, '2019-07-07', 4),
(123, 'Nikhil Sethi', 'M', 69000, '2023-01-05', 4),
(124, 'Tanya Arora', 'F', 77000, '2021-12-18', 4),
(125, 'Dev Malhotra', 'M', 98000, '2018-05-20', 5),
(126, 'Aditi Mishra', 'F', 91000, '2020-03-16', 5),
(127, 'Varun Nair', 'M', 87000, '2021-10-08', 5),
(128, 'Shreya Das', 'F', 93000, '2019-11-25', 5),
(129, 'Yash Thakur', 'M', 79000, '2022-07-14', 5),
(130, 'Diya Kapoor', 'F', 96000, '2023-02-02', 5);
```

**INSERTING 8 PROJECTS**

```
INSERT INTO Project
(project_id, project_name, budget, dept_id)
VALUES
(201, 'AI Chatbot', 500000, 1),
(202, 'Cloud Migration', 750000, 1),
(203, 'Employee Training', 250000, 2),
(204, 'Financial Analysis', 600000, 3),
(205, 'Digital Marketing', 450000, 4),
(206, 'Market Research', 350000, 4),
(207, 'Machine Learning Research', 900000, 5),
(208, 'AI Security System', 1000000, 5);
```

**ASSIGNING EMPLOYEES TO PROJECTS**

```
INSERT INTO Employee_Project
(emp_id, project_id, hours_worked)
VALUES
(101, 201, 120),
(102, 201, 100),
(103, 202, 150),
(104, 202, 110),
(105, 201, 130),
(106, 202, 90),
(107, 203, 100),
(108, 203, 120),
(109, 203, 80),
(110, 203, 110),
(111, 203, 90),
(112, 203, 130),
(113, 204, 140),
(114, 204, 150),
(115, 204, 120),
(116, 204, 135),
```

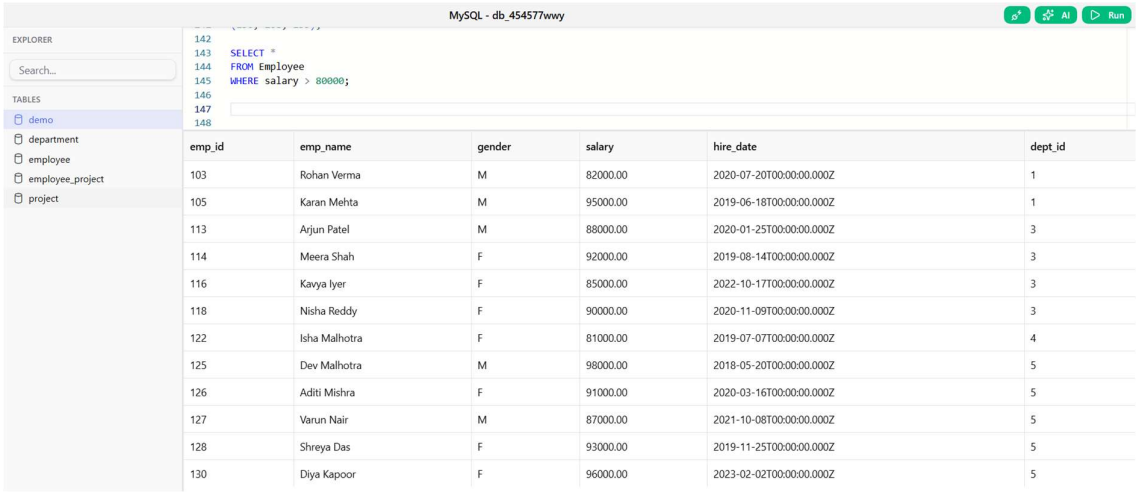
```
(117, 204, 100),
(118, 204, 145),
(119, 205, 110),
(120, 205, 100),
(121, 206, 120),
(122, 205, 140),
(123, 206, 90),
(124, 206, 130),
(125, 207, 160),
(126, 207, 140),
(127, 208, 150),
(128, 208, 160),
(129, 207, 120),
(130, 208, 135);
```

## SQL QUERIES:

### SELECTION

```
SELECT *
FROM Employee
WHERE salary > 80000;
```

### OUTPUT:

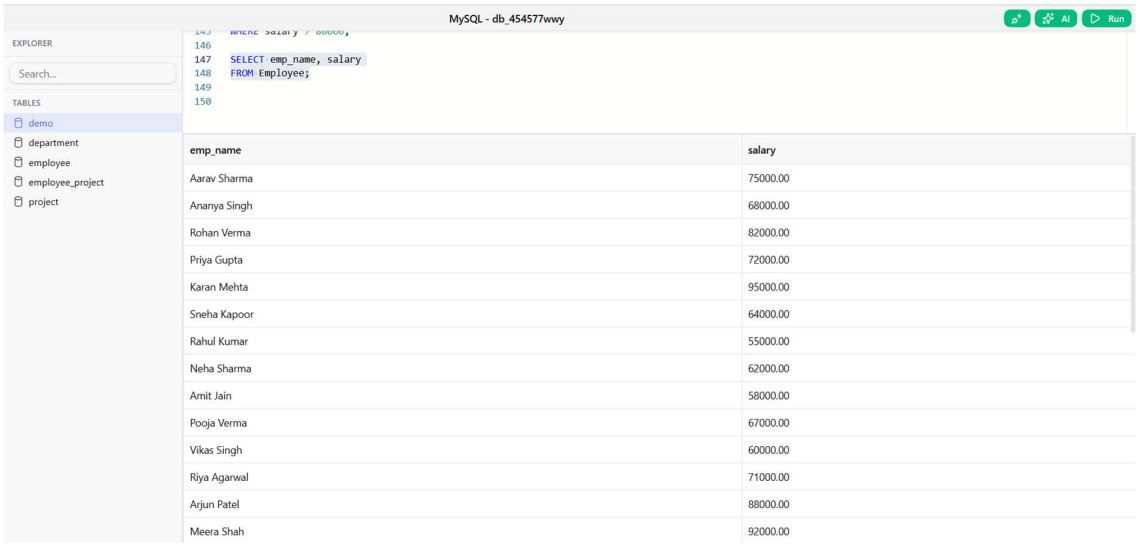



| emp_id | emp_name | gender | salary | hire_date | dept_id |
|---|---|---|---|---|---|
| 103 | Rohan Verma | M | 82000.00 | 2020-07-20T00:00:00.000Z | 1 |
| 105 | Karan Mehta | M | 95000.00 | 2019-06-18T00:00:00.000Z | 1 |
| 113 | Arjun Patel | M | 88000.00 | 2020-01-25T00:00:00.000Z | 3 |
| 114 | Meera Shah | F | 92000.00 | 2019-08-14T00:00:00.000Z | 3 |
| 116 | Kavya Iyer | F | 85000.00 | 2022-10-17T00:00:00.000Z | 3 |
| 118 | Nisha Reddy | F | 90000.00 | 2020-11-09T00:00:00.000Z | 3 |
| 122 | Isha Malhotra | F | 81000.00 | 2019-07-07T00:00:00.000Z | 4 |
| 125 | Dev Malhotra | M | 98000.00 | 2018-05-20T00:00:00.000Z | 5 |
| 126 | Aditi Mishra | F | 91000.00 | 2020-03-16T00:00:00.000Z | 5 |
| 127 | Varun Nair | M | 87000.00 | 2021-10-08T00:00:00.000Z | 5 |
| 128 | Shreya Das | F | 93000.00 | 2019-11-25T00:00:00.000Z | 5 |
| 130 | Diya Kapoor | F | 96000.00 | 2023-02-02T00:00:00.000Z | 5 |

## PROJECTION

```
SELECT emp_name, salary
FROM Employee;
```

## OUTPUT:




| emp_name | salary |
|---|---|
| Aarav Sharma | 75000.00 |
| Ananya Singh | 68000.00 |
| Rohan Verma | 82000.00 |
| Priya Gupta | 72000.00 |
| Karan Mehta | 95000.00 |
| Sneha Kapoor | 64000.00 |
| Rahul Kumar | 55000.00 |
| Neha Sharma | 62000.00 |
| Amit Jain | 58000.00 |
| Pooja Verma | 67000.00 |
| Vikas Singh | 60000.00 |
| Riya Agarwal | 71000.00 |
| Arjun Patel | 88000.00 |
| Meera Shah | 92000.00 |

## AGGREGATE FUNCTIONS

```
SELECT
    COUNT(*) AS total_employees,
    SUM(salary) AS total_salary,
    AVG(salary) AS average_salary,
    MAX(salary) AS highest_salary,
    MIN(salary) AS lowest_salary
FROM Employee;
```

## OUTPUT:

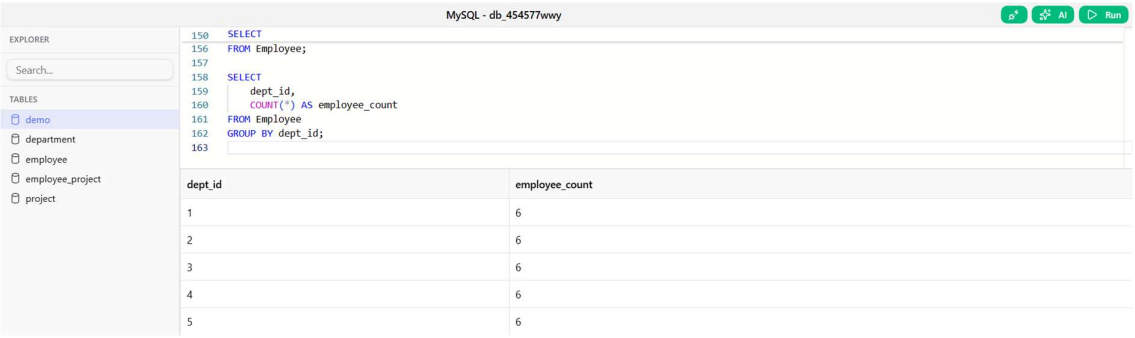



| total_employees | total_salary | average_salary | highest_salary | lowest_salary |
|---|---|---|---|---|
| 30 | 2314000.00 | 77133.333333 | 98000.00 | 55000.00 |

## GROUP BY

```
SELECT
    dept_id,
    COUNT(*) AS employee_count
FROM Employee
GROUP BY dept_id;
```

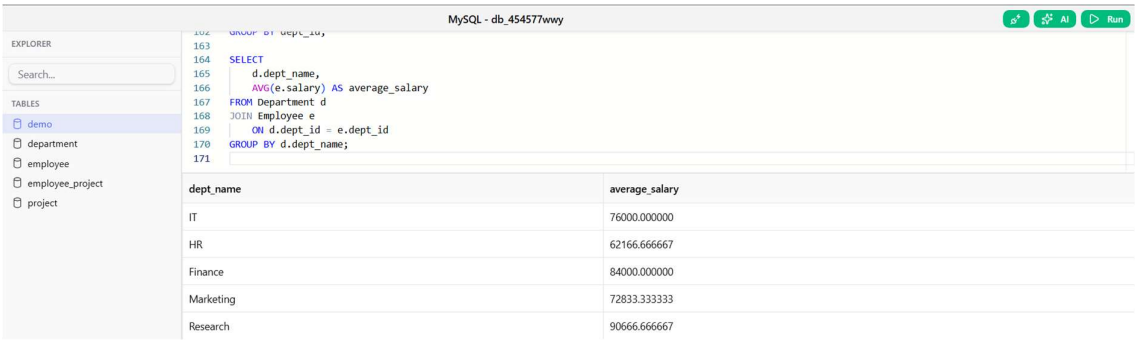
## OUTPUT:




## GROUP BY WITH JOIN

```
SELECT
    d.dept_name,
    AVG(e.salary) AS average_salary
FROM Department d
JOIN Employee e
    ON d.dept_id = e.dept_id
GROUP BY d.dept_name;
```

## OUTPUT:



## HAVING

```
SELECT
    d.dept_name,
    AVG(e.salary) AS average_salary
FROM Department d
JOIN Employee e
    ON d.dept_id = e.dept_id
GROUP BY d.dept_name
HAVING AVG(e.salary) > 75000;
```

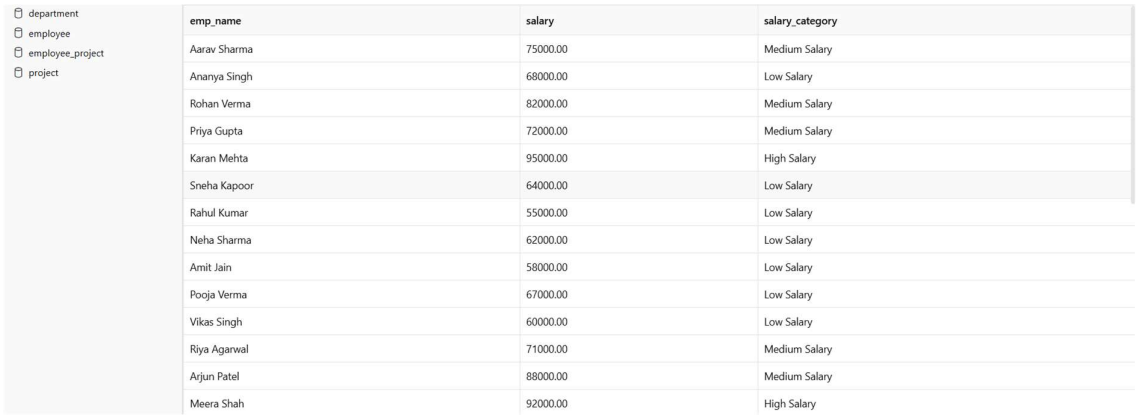
**OUTPUT**:




## CASE EXPRESSIONS

```
SELECT
    emp_name,
    salary,
    CASE
        WHEN salary >= 90000 THEN 'High Salary'
        WHEN salary >= 70000 THEN 'Medium Salary'
        ELSE 'Low Salary'
    END AS salary_category
FROM Employee;
```

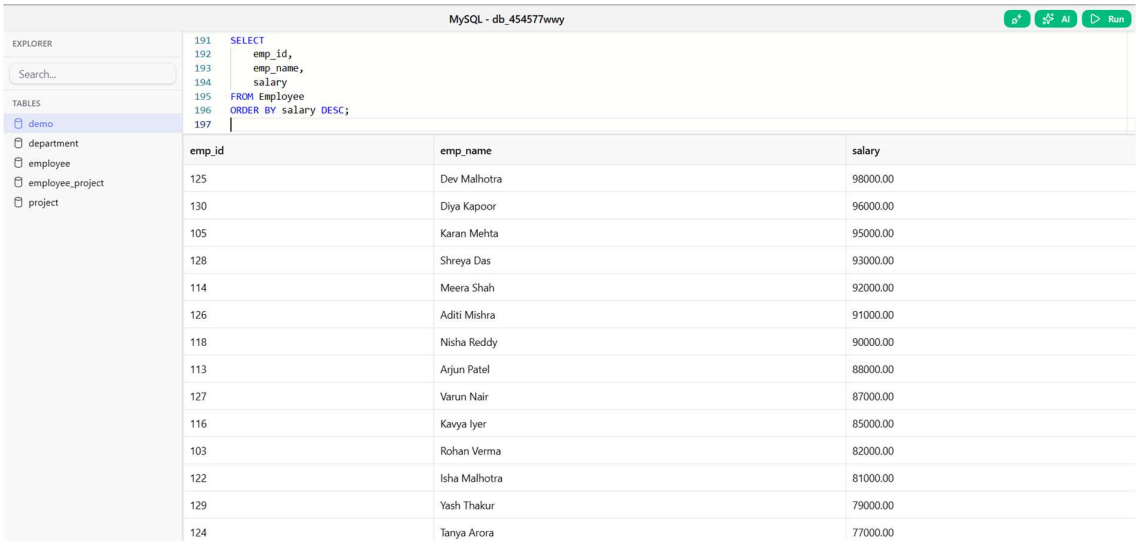
**OUTPUT**:



**ORDER BY**

```
SELECT
    emp_id,
    emp_name,
    salary
FROM Employee
ORDER BY salary DESC;
```

**OUTPUT:**

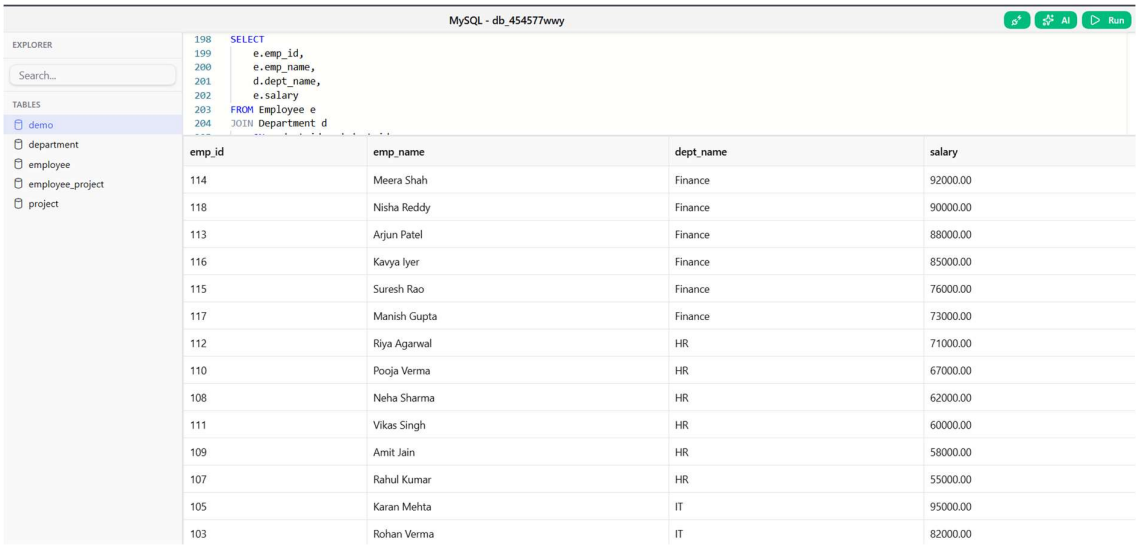



| emp_id | emp_name | salary |
|---|---|---|
| 125 | Dev Malhotra | 98000.00 |
| 130 | Diya Kapoor | 96000.00 |
| 105 | Karan Mehta | 95000.00 |
| 128 | Shreya Das | 93000.00 |
| 114 | Meera Shah | 92000.00 |
| 126 | Aditi Mishra | 91000.00 |
| 118 | Nisha Reddy | 90000.00 |
| 113 | Arjun Patel | 88000.00 |
| 127 | Varun Nair | 87000.00 |
| 116 | Kavya Iyer | 85000.00 |
| 103 | Rohan Verma | 82000.00 |
| 122 | Isha Malhotra | 81000.00 |
| 129 | Yash Thakur | 79000.00 |
| 124 | Tanya Arora | 77000.00 |

**COMPLETE EMPLOYEE-DEPARTMENT REPORT**

```
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name,
    e.salary
FROM Employee e
JOIN Department d
    ON e.dept_id = d.dept_id
ORDER BY d.dept_name, e.salary DESC;
```

**OUTPUT**:

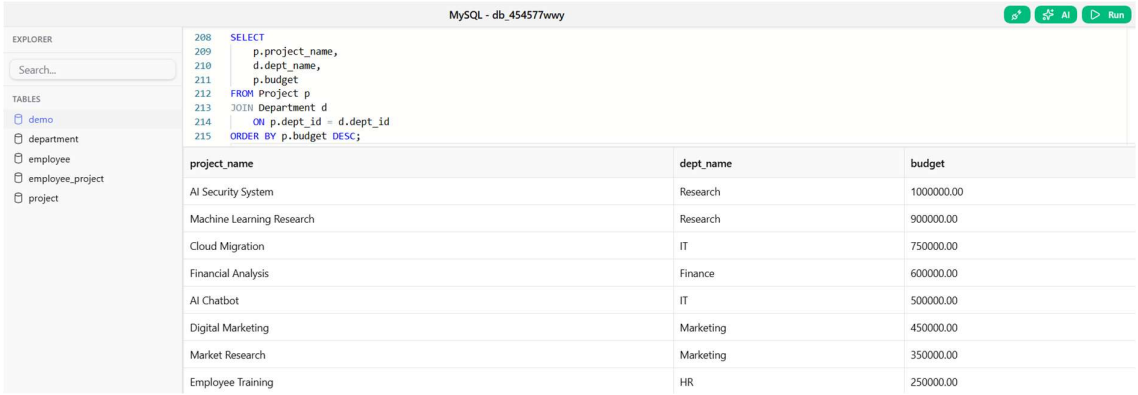



| emp_id | emp_name | dept_name | salary |
|---|---|---|---|
| 114 | Meera Shah | Finance | 92000.00 |
| 118 | Nisha Reddy | Finance | 90000.00 |
| 113 | Arjun Patel | Finance | 88000.00 |
| 116 | Kavya Iyer | Finance | 85000.00 |
| 115 | Suresh Rao | Finance | 76000.00 |
| 117 | Manish Gupta | Finance | 73000.00 |
| 112 | Riya Agarwal | HR | 71000.00 |
| 110 | Pooja Verma | HR | 67000.00 |
| 108 | Neha Sharma | HR | 62000.00 |
| 111 | Vikas Singh | HR | 60000.00 |
| 109 | Amit Jain | HR | 58000.00 |
| 107 | Rahul Kumar | HR | 55000.00 |
| 105 | Karan Mehta | IT | 95000.00 |
| 103 | Rohan Verma | IT | 82000.00 |

**PROJECT DETAILS**

```
SELECT
    p.project_name,
    d.dept_name,
    p.budget
FROM Project p
JOIN Department d
    ON p.dept_id = d.dept_id
ORDER BY p.budget DESC;
```

**OUTPUT**:

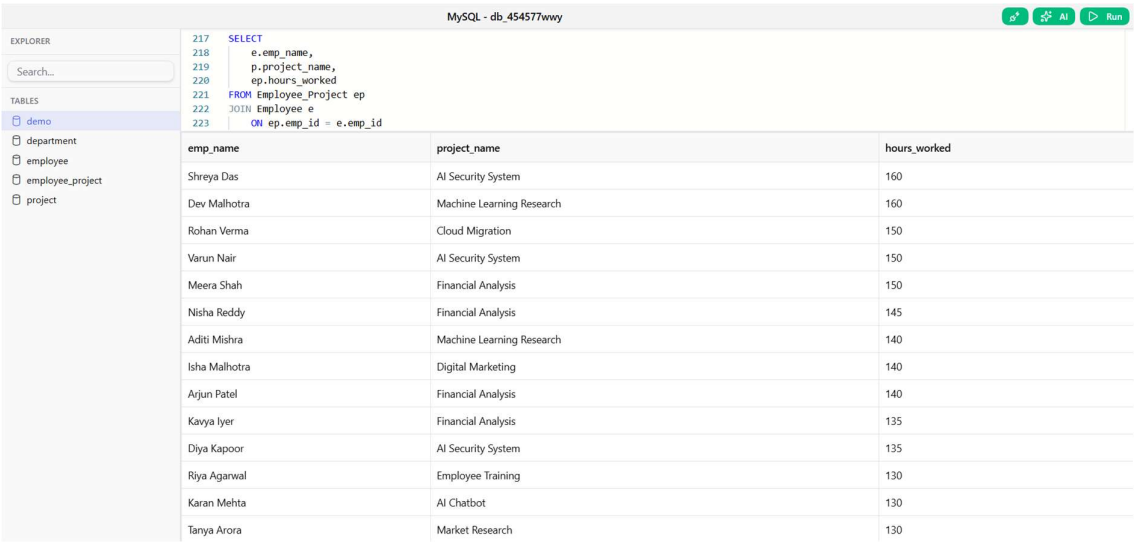



| project_name | dept_name | budget |
|---|---|---|
| AI Security System | Research | 1000000.00 |
| Machine Learning Research | Research | 900000.00 |
| Cloud Migration | IT | 750000.00 |
| Financial Analysis | Finance | 600000.00 |
| AI Chatbot | IT | 500000.00 |
| Digital Marketing | Marketing | 450000.00 |
| Market Research | Marketing | 350000.00 |
| Employee Training | HR | 250000.00 |

## EMPLOYEES WORKING ON PROJECTS

```
SELECT
    e.emp_name,
    p.project_name,
    ep.hours_worked
FROM Employee_Project ep
JOIN Employee e
    ON ep.emp_id = e.emp_id
JOIN Project p
    ON ep.project_id = p.project_id
ORDER BY ep.hours_worked DESC;
```

## OUTPUT:




| emp_name | project_name | hours_worked |
|---|---|---|
| Shreya Das | AI Security System | 160 |
| Dev Malhotra | Machine Learning Research | 160 |
| Rohan Verma | Cloud Migration | 150 |
| Varun Nair | AI Security System | 150 |
| Meera Shah | Financial Analysis | 150 |
| Nisha Reddy | Financial Analysis | 145 |
| Aditi Mishra | Machine Learning Research | 140 |
| Isha Malhotra | Digital Marketing | 140 |
| Arjun Patel | Financial Analysis | 140 |
| Kavya Iyer | Financial Analysis | 135 |
| Diya Kapoor | AI Security System | 135 |
| Riya Agarwal | Employee Training | 130 |
| Karan Mehta | AI Chatbot | 130 |
| Tanya Arora | Market Research | 130 |